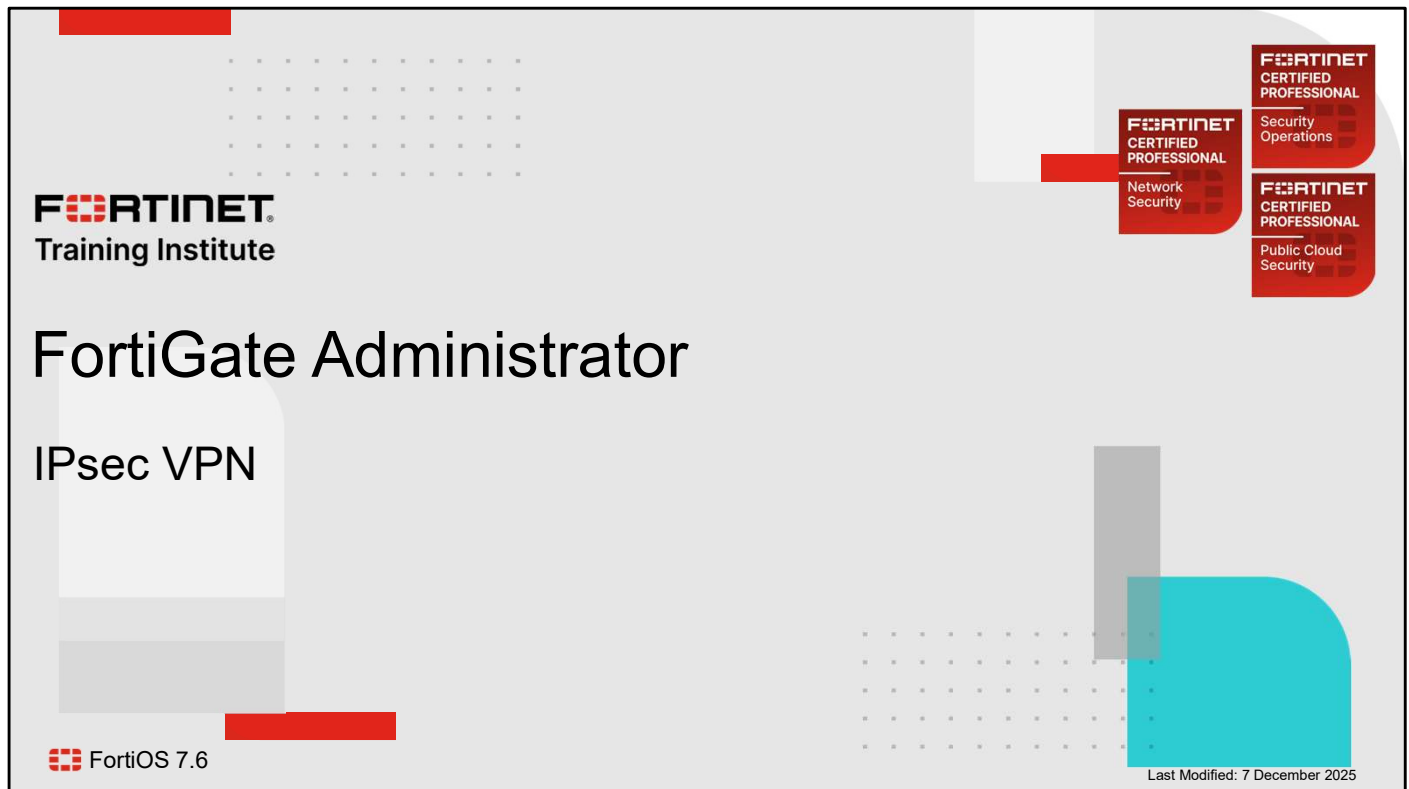
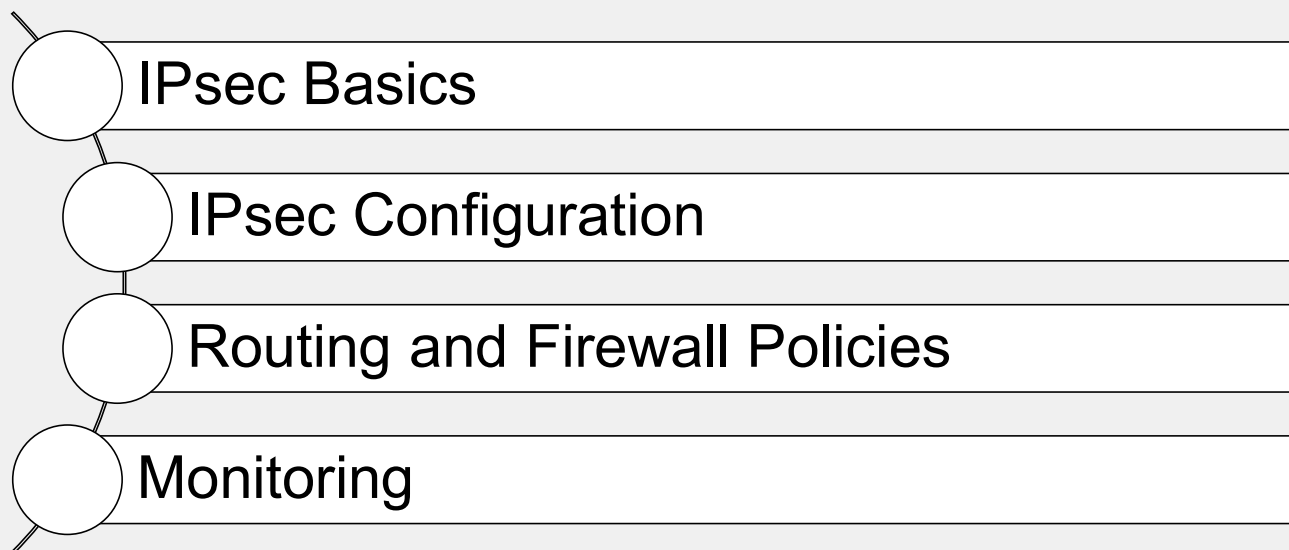




In this lesson, you will learn about the architectural components of IPsec VPN and how to configure them.

Lesson Overview



In this lesson, you will learn about the topics shown on this slide.

IPsec Basics

Objectives

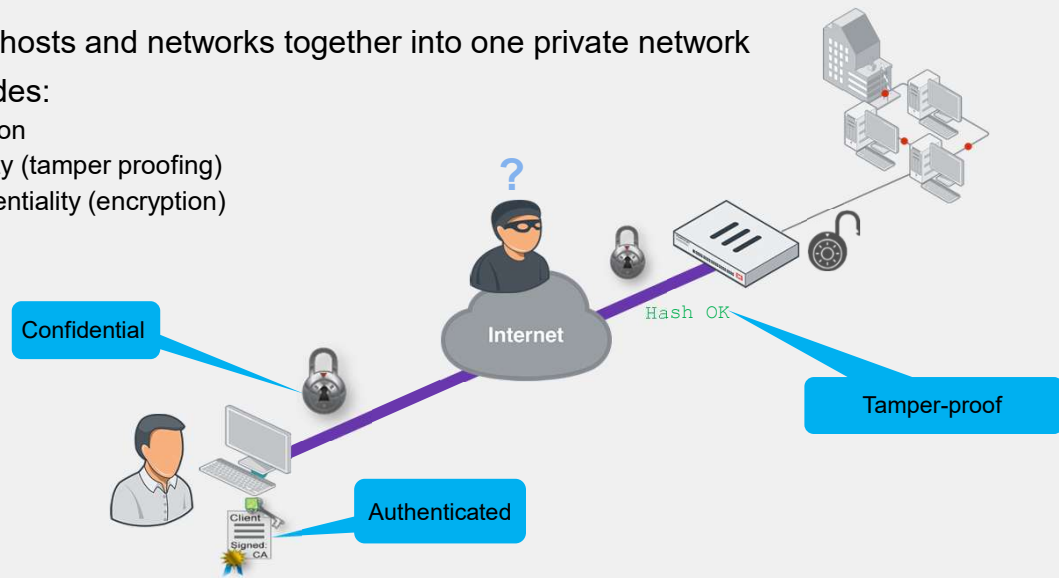
- Describe the benefits of IPsec VPN
- Be familiar with the IPsec protocol
- Understand how IPsec works
- Select an appropriate VPN topology

After completing this section, you should be able to achieve the objectives shown on this slide.

By demonstrating competence in IPsec basics, you will be able to understand IPsec concepts and benefits.

What Is IPsec?

- Joins remote hosts and networks together into one private network
- Usually provides:
 - Authentication
 - Data integrity (tamper proofing)
 - Data confidentiality (encryption)



What is IPsec? When should you use it?

IPsec is a vendor-neutral set of standard protocols that is used to join two physically distinct LANs. The LANs are joined as if they were a single logical network, despite being separated by the internet.

In theory, IPsec *does* support null encryption—that is, you can make VPNs that don't encrypt traffic. IPsec also supports null data integrity. But does that provide any advantages over plain traffic? No. No one can trust traffic that may have had an attack injected by an attacker. Rarely do people want data sent by an unknown source. Most people also want private network data, such as credit card transactions and medical records, to remain private.

Regardless of the vendor, IPsec VPNs almost always have settings that allow them to provide three important benefits:

- Authentication: to verify the identity of both ends
- Data integrity (or HMAC): to prove that encapsulated data has not been tampered with as it crosses a potentially hostile network
- Confidentiality (or encryption): to make sure that only the intended recipient can read the message

What Is the IPsec Protocol?

- Multiple protocols that work together
 - Authentication Header (AH) provides integrity but not encryption
 - AH is defined in the RFC, but FortiGate does not use it
- Port numbers and encapsulation vary by network address translation (NAT)

Protocol	NAT Traversal (NAT-T)	No NAT
IKE RFC 2409 (IKEv1) RFC 4306 (IKEv2)	IP protocol 17: UDP port 500 (UDP 4500 for rekey, quick mode, mode-cfg)	IP protocol 17: UDP port 500
ESP RFC 4303	IP protocol 17: UDP port 4500 (encapsulated)	IP protocol 50

- If required, set a custom port for both IKE and IKE NAT-T (initiator and responder)*:

```
config system settings
    set ike-port <port>
end
```

* Custom port range: 1024–65535. FortiGate always listens on UDP port 4500 (responder only)

If you're passing your VPN through firewalls, it helps to know which protocols to allow.

IPsec is a suite of separate protocols, which includes:

- Internet Key Exchange (IKE): used to authenticate peers, exchange keys, and negotiate the encryption and checksums that are used—essentially, it is the *control channel*
- AH: contains the authentication header—the checksums that verify the integrity of the data
- Encapsulating Security Payload (ESP): the encapsulated security payload—the encrypted payload, which is essentially the *data channel*

So, if you must pass IPsec traffic through a firewall, remember that allowing only one protocol or port number is usually not enough.

Note that the IPsec RFC mentions AH, however, AH does not offer encryption, which is an important benefit. Therefore, FortiGate does not use AH. As a result, you don't need to allow the AH IP protocol (51).

To set up a VPN, you must configure matching settings on both ends of the VPN—whether the VPN is between two FortiGate devices, FortiGate and FortiClient, or a third-party device and FortiGate. If the settings don't match, the tunnel setup fails.

The default ports for standard IKE traffic and IKE NAT-T traffic are UDP 500 and UDP 4500, respectively. You can use the CLI command shown on this slide to configure a custom port for both IKE and IKE NAT-T. The custom port is used to initiate and respond to tunnel requests. If NAT is detected, then the custom port can be used for both IKE and UDP-encapsulated ESP traffic. Note that FortiGate always listens for port UDP 4500 regardless of the custom port settings. This enables FortiGate to negotiate NAT-T tunnels on custom and standard ports.

How Does IPsec Work?

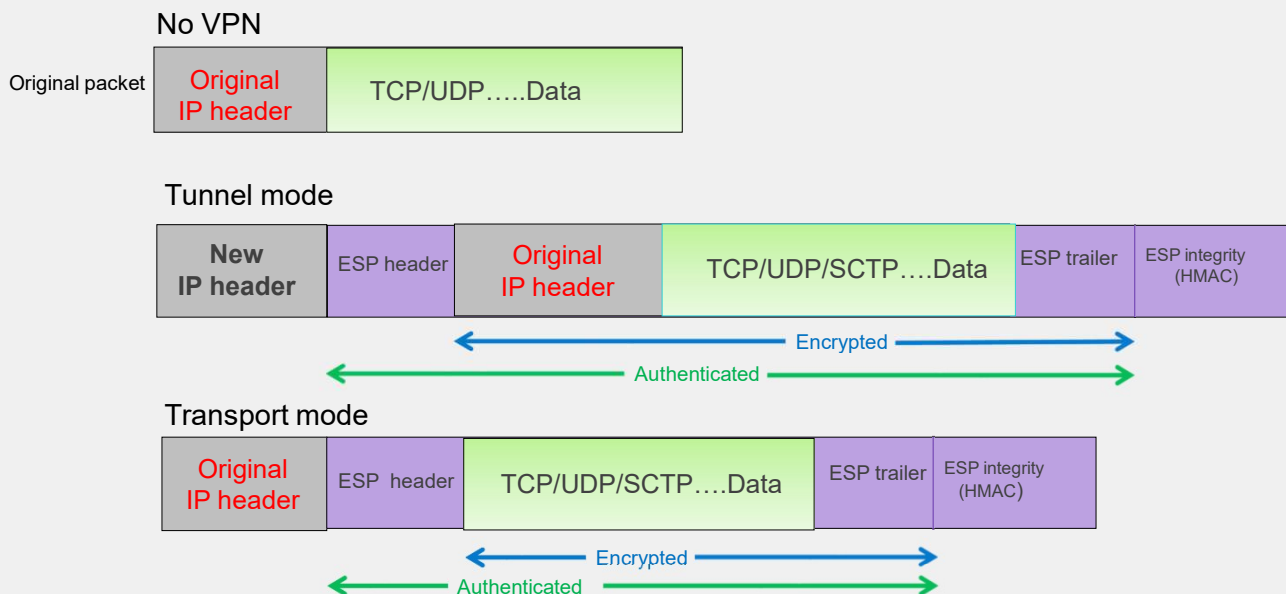
- Encapsulation
 - Other protocols wrapped inside IPsec
 - What's inside? Varies by mode:
 - Transport mode—TCP/UDP
 - Tunnel mode—additional IP layer, then TCP/UDP
- Negotiation
 - Authentication
 - Handshake to exchange keys, settings



IPsec provides services at the IP (network) layer. During tunnel establishment, both ends negotiate the encryption and authentication algorithms to use.

After the tunnel has been negotiated and is up, data is encrypted and encapsulated into ESP packets.

ESP Encapsulation—Tunnel or Transport Mode



What's encapsulated? It depends on the encapsulation mode IPsec uses. IPsec can operate in two modes: transport mode and tunnel mode.

- Transport mode directly encapsulates and protects the fourth layer (transport) and above. It does not protect the original IP header and does not add an additional IP header.
- Tunnel mode is a true tunnel. It encapsulates the whole IP packet and adds a new IP header at the beginning. After the IPsec packet reaches the remote LAN and is unwrapped, the original packet can continue on its journey.

Note that after you remove the VPN-related headers, a transport mode packet can't be transmitted any further; it has no second IP header inside, so it's not routable. For that reason, this mode is usually used only for end-to-end (or client-to-client) VPNs.

What Is IKE?

- Default ports: UDP port 500 (and UDP port 4500 when crossing NAT)
- Negotiates a tunnel's private keys, authentication, and encryption
- Phases:
 - Phase 1
 - Phase 2
- Versions
 - IKEv1 (legacy, wider adoption)
 - IKEv2 (new, simpler operation)

IKE uses UDP port 500. If NAT-T is enabled in a NAT scenario, IKE uses UDP port 4500.

IKE establishes an IPsec VPN tunnel. FortiGate uses IKE to negotiate with the peer and determine the IPsec security association (SA). The IPsec SA defines the authentication, keys, and settings that FortiGate uses to encrypt and decrypt that peer's packets. It is based on the Internet Security Association and Key Management Protocol (ISAKMP).

IKE defines two phases: phase 1 and phase 2.

There are two IKE versions: IKEv1 and IKEv2. Even though IKEv2 is a newer version and features a simpler protocol operation, this lesson focuses on IKEv1 only, because of its much wider adoption.

IKEv1 vs. IKEv2

Feature	IKEv1	IKEv2
Exchange modes	- Main - Total messages: 9 (6 for phase 1, 3 for phase 2) - Aggressive - Total messages: 6 (3 for phase 1, 3 for phase 2)	- One exchange procedure only - Total messages: 4 (one child SA only)
Authentication methods	Symmetric: - Pre-shared key (PSK) - Certificate signature - Extended authentication (XAuth)	Asymmetric: - PSK - Certificate signature - EAP (pass-through—no client support)
NAT-T	Supported as extension	Native support
Reliability	Unreliable—messages are not acknowledged	Reliable—messages are acknowledged
Dial-up phase 1 matching by ID	- Peer ID + aggressive mode + PSK - Peer ID + main mode + certificate signature	- Peer ID - Network ID
Traffic selector narrowing	Not supported	Supported

This slide shows a table comparing some of the IKEv1 and IKEv2 features that FortiOS supports. IKEv2 provides a simpler operation, which is the result of using a single exchange mode and requiring less messages to bring up the tunnel.

Authentication-wise, both versions support PSK and certificate signature. Although only IKEv1 supports XAuth, IKEv2 supports EAP, which is equivalent to XAuth. However, the FortiOS IKEv2 EAP implementation is pass-through only. That is, FortiOS doesn't support EAP as a client, which means that you cannot revoke access to peers using IKEv2 unless you use a certificate signature. With IKEv1, you can deny access to VPN peers without having to use a certificate signature by using XAuth. IKEv2 also supports asymmetric authentication, which enables you to configure each peer to use a different authentication method.

Both IKE versions support NAT-T. However, IKEv2 supports NAT-T natively, while IKEv1 supports NAT-T as an extension. Also, IKEv2 is a more reliable protocol than IKEv1 because, like TCP, peers must acknowledge the messages exchanged between them. IKEv1 doesn't support such a mechanism.

When you configure multiple dial-up IPsec VPNs, IKEv2 makes it simpler to match the intended gateway by peer ID. With IKEv2, you can either use the standard peer ID attribute or the Fortinet proprietary network ID attribute to indicate the phase 1 gateway to match on the dial-up server, regardless of the authentication mode in use. However, with IKEv1, you can use the peer ID only, and then combine it with aggressive mode and pre-shared key authentication, or with main mode and certificate signature authentication.

Finally, IKEv2 allows the responder to choose a subset of the traffic the initiator proposes. This is called traffic selector narrowing and enables you to have more flexible phase 2 selector configurations. Traffic selector narrowing enables a peer to automatically narrow down its traffic selector addresses, so it agrees with the traffic selector the remote peer proposes.

Negotiation—Security Association (SA)

- IKE allows the parties involved in a transaction to set up their Security Associations (SAs)
 - SAs are the basis for building security functions into IPsec
 - In normal two-way traffic, the exchange is secured by a pair of SAs
 - IPsec administrators decide the encryption and authentication algorithms that can be used in the exchange
- IKE uses two distinct phases:
 - Phase 1 → Outcome: IKE SA
 - Phase 2 → Outcome: IPsec SA

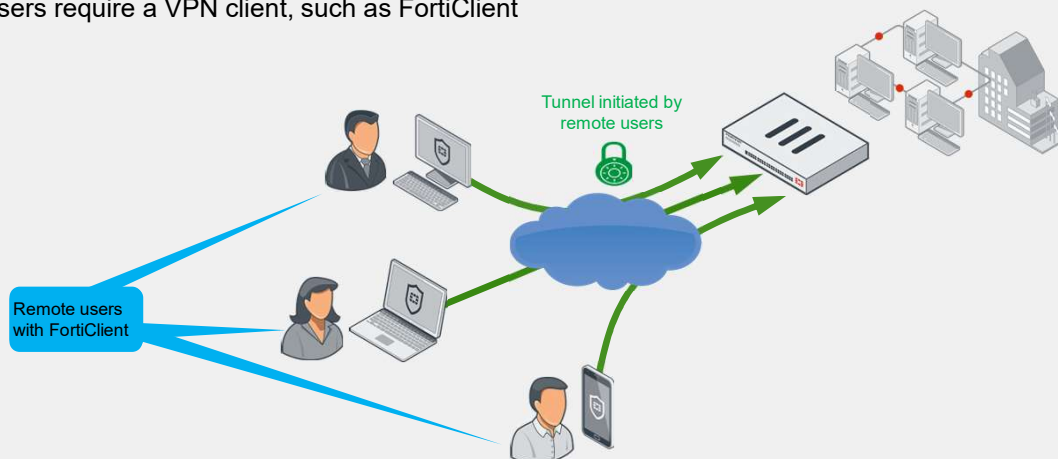
In order to create an IPsec tunnel, both devices must establish their SAs and secret keys, which are facilitated by the IKE protocol.

The IPsec architecture uses SAs as the basis for building security functions into IPsec. An SA is the bundle of algorithms and parameters being used to encrypt and authenticate data travelling through the tunnel. In normal two-way traffic, this exchange is secured by a pair of SAs, one for each traffic direction. Essentially, both sides of the tunnel must agree on the security rules. If both sides cannot agree on the rules for sending data and verifying each other's identity, then the tunnel is not established. SAs expire and need to be renegotiated by the peers after they have reached their lifetime.

IKE uses two distinct phases: phase 1 and phase 2. Each phase negotiates different SA types. The SA negotiated during phase 1 is called IKE SA, and the SA negotiated during phase 2 is called IPsec SA. FortiGate uses IKE SAs for setting up a secure channel to negotiate IPsec SAs. FortiGate uses IPsec SAs for encrypting and decrypting the data sent and received, respectively, through the tunnel.

VPN Topologies—Remote Access

- Remote users connect to corporate resources
 - FortiGate is configured as dial-up server—only clients can initiate the VPN
 - Users require a VPN client, such as FortiClient



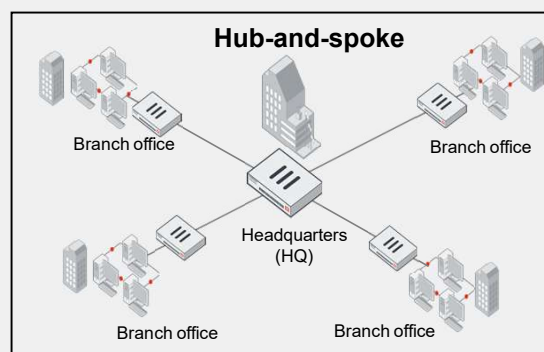
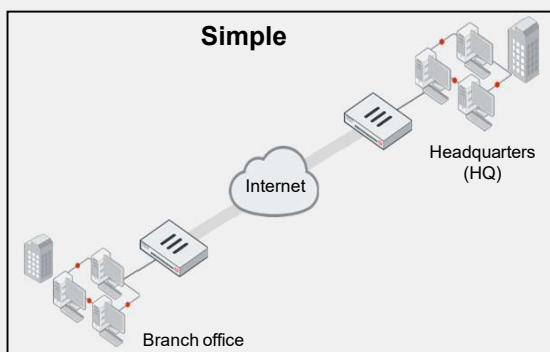
Use remote access VPNs when remote internet users need to securely connect to the office to access corporate resources. The remote user connects to a VPN server located on the corporate premises, such as FortiGate, to establish a secure tunnel. After the user is authenticated, FortiGate provides access to network resources, based on the permissions granted to that user.

In a remote access VPN, FortiGate is usually configured as a dial-up server. You will learn more about dial-up VPNs in this lesson. The IP address of the remote internet user is usually dynamic. Because FortiGate does not know the IP address of the remote user, only the remote user can initiate a VPN connection request.

The remote user side needs a VPN client, such as FortiClient. You must configure FortiClient to match the VPN server settings. FortiClient takes care of establishing the tunnel, as well as routing the traffic destined to the remote site through the tunnel.

In addition, you can use one remote access VPN configuration on your FortiGate device for many remote users. FortiGate establishes a separate tunnel for each of them.

VPN Topologies—Site-to-Site

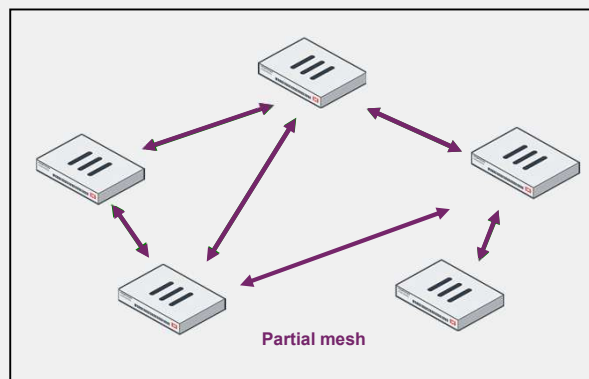
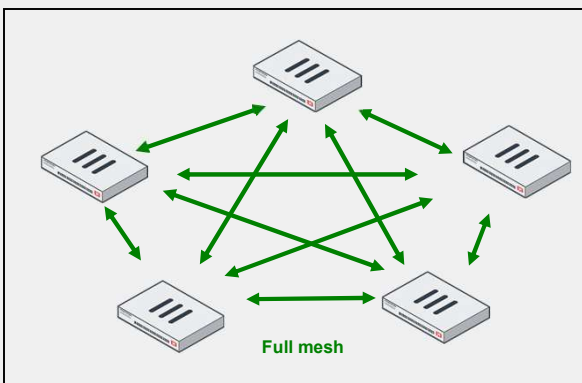


Site-to-site VPN is also known as LAN-to-LAN VPN. A simple site-to-site deployment involves two peers communicating directly to connect two networks located at different offices.

When you need to connect more than two locations, you can use a hub-and-spoke topology. In hub-and-spoke, all clients connect through a central hub. In the example shown on this slide, the clients—spokes—are branch office FortiGate devices. For any branch office to reach another branch office, its traffic must pass through the hub. One advantage of this topology is that the configuration needed is easy to manage. Another advantage is that only the FortiGate at HQ must be very powerful because it handles all tunnels simultaneously, while the branch office FortiGate devices require much fewer resources because they maintain only one tunnel. One disadvantage is that communication between branch offices through HQ is slower than in a direct connection, especially if your HQ is physically distant. Also, if the FortiGate device at HQ fails, VPN failure is company-wide.

VPN Topologies—Site-to-Site (Contd)

Full mesh and partial mesh



In a mesh topology, you can connect FortiGate devices directly and therefore bypass HQ. Two variations of mesh topology exist: full mesh and partial mesh. Full mesh connects every location to every other location. The higher the number of FortiGate devices, the higher the number of tunnels to configure on each FortiGate device. For example, in a topology with five FortiGate devices, you would need to configure four tunnels on each device, for a total of 20 tunnels. This topology causes less latency and requires much less HQ bandwidth than hub-and-spoke, but requires each FortiGate device to be more powerful. Partial mesh attempts to compromise, minimizing required resources but also latency. Partial mesh can be appropriate if communication is not required between every location. However, the configuration of each FortiGate device is more complex than in hub-and-spoke. Routing, especially, may require extensive planning.

Generally, the more locations you have, hub-and-spoke will be cheaper, but slower, than a mesh topology. Mesh places less strain on the central location. It's more fault-tolerant, but also more expensive.

VPN Topologies—Comparison

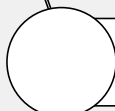
Hub-and-Spoke	Partial Mesh	Full Mesh
Easy configuration	Moderate configuration	Complex configuration
Few tunnels	Medium number of tunnels	Many tunnels
High central bandwidth	Medium bandwidth in hub sites	Low bandwidth
Not fault tolerant	Some fault tolerance	Fault tolerant
Low system requirements on average, but high for center	Medium system requirements	High system requirements
Scalable	Somewhat scalable	Difficult to scale
No direct communication between spokes	Direct communication between some sites	Direct communication between all sites

To review, this slide shows a high-level comparison of VPN topologies. You should choose the topology that is most appropriate to your situation.

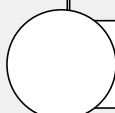
Lesson Progress



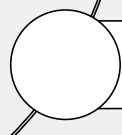
IPsec Basics



IPsec Configuration



Routing and Firewall Policies



Monitoring

Good job! You have now been introduced to IPsec.

Now, you will learn about IPsec configuration.

IPsec Configuration

Objectives

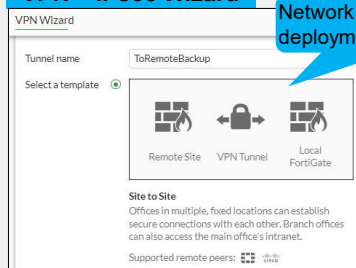
- Configure IPsec VPN using the IPsec wizard
- Configure IPsec VPN manually

After completing this section, you should be able to achieve the objectives shown on this slide.

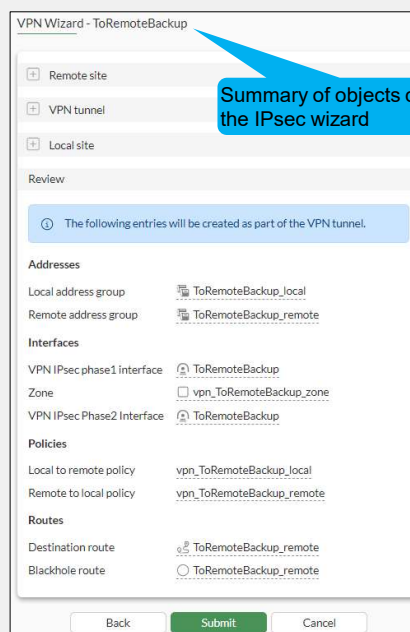
By demonstrating competence in IPsec configuration, you will also be able to successfully determine the settings required for your IPsec VPN deployment.

IPsec Wizard

VPN > IPsec Wizard



Network diagram describing deployment type



Summary of objects created by the IPsec wizard

When you create an IPsec tunnel on the GUI, FortiGate redirects you to the **IPsec Wizard**. The wizard simplifies the creation of the new VPN by walking you through a four to five-step process. The first step is to select a template type. If you want to manually configure your VPN, you can select **Custom** as **Template type**, upon which FortiGate takes you directly to the phase 1 and phase 2 settings of the new VPN.

If you want the wizard to configure the VPN for you, then select the template type (**Site to Site**, **Hub-and-Spoke**, or **Remote Access**) that best matches your VPN. After that, the wizard asks you for key information, such as the remote gateway information, authentication method, interfaces involved, and subnets. Based on the input you provide, the wizard applies one of the preconfigured IPsec tunnel templates comprising IPsec phase 1 and 2 settings and other related firewall address objects, routing settings, and firewall policies needed for the new tunnel to work.

In addition, the wizard shows a network diagram that changes based on the input you provide. The purpose of the diagram is for the administrator to have a visual understanding of the IPsec VPN deployment that the wizard configures based on the input it receives.

At the end of the wizard, the wizard provides a summary of the configuration changes made in the system, and that the administrator can review if needed.

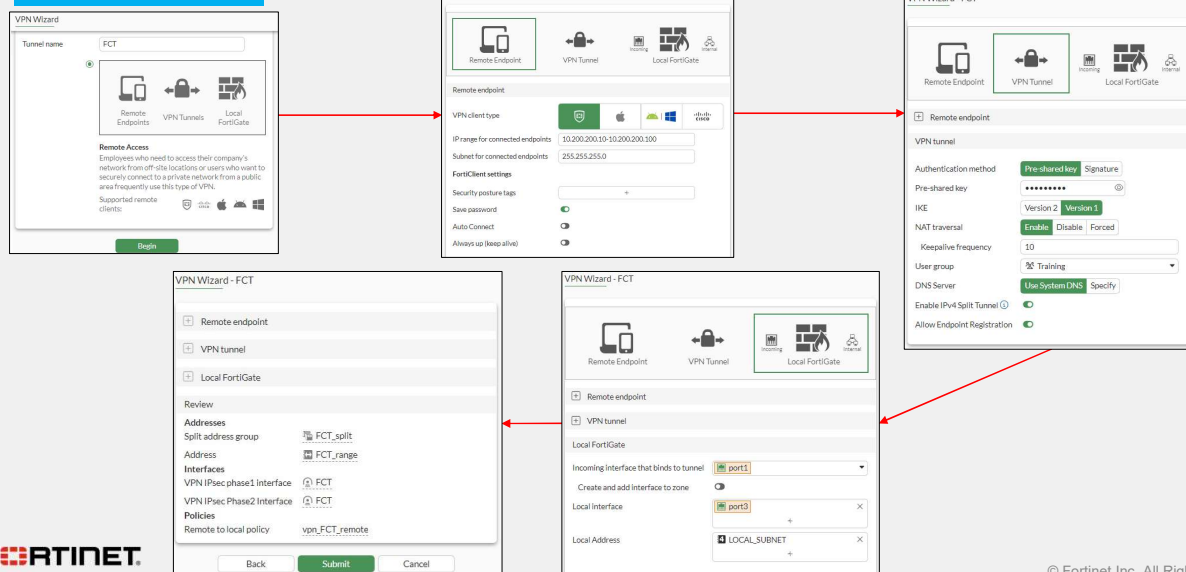
If you are new to FortiGate, or don't have much experience with IPsec VPNs, using the IPsec wizard is recommended. You can later adjust the configuration applied by the wizard to match your specific needs.

Note that, in this lesson, you will learn only about IKEv1 configuration.

Using the IPsec Wizard for a FortiClient VPN

- Simplifies IPsec configuration for a FortiClient VPN

VPN > IPsec Wizard



A common use of the IPsec wizard is for configuring a remote access VPN for FortiClient users. The wizard enables IKE mode config, XAuth, and other appropriate settings for FortiClient users. You will learn more about IKE mode config and XAuth in this lesson.

The images on this slide show the four-step process used by the IPsec wizard for assisting the administrator on the FortiClient VPN configuration.

Phase 1—Overview

- Each peer of the tunnel—the initiator and the responder—connects and begins to set up the VPN
- On the first connection, the channel is not secure
 - Unencrypted keys can be intercepted
- To exchange sensitive private keys, both peers create a secure channel
 - Both peers negotiate the real keys for the tunnel later

Phase 1 takes place when each peer of the tunnel—the initiator and the responder—connects and begins to set up the VPN. The initiator is the peer that starts the phase 1 negotiation, while the responder is the peer that responds to the initiator request.

When the peers first connect, the channel is not secure. An attacker in the middle could intercept unencrypted keys. Neither peer has a strong guarantee of the other peer's identity, so how can they exchange sensitive private keys? They can't. First, both peers create a secure tunnel. Then, they use this secure tunnel to negotiate the real keys for the tunnel later.

Phase 1—How it Works

1. Authenticate peers
 - PSK or digital signature
 - XAuth
2. Negotiate one bidirectional SA (called IKE SA)
 - In IKE v1, two possible ways:
 - Main mode
 - Aggressive mode
 - Not the same as IPsec SA
 - Encrypted tunnel for Diffie-Hellman (DH)

Bidirectional SA: same key to encrypt the outgoing traffic and decrypt the incoming traffic



3. DH exchange for secret keys

Now you'll examine how phase 1 works.

The purpose of phase 1 is to authenticate peers and set up a secure channel for negotiating the phase 2 SAs (or IPsec SAs) that are later used to encrypt and decrypt traffic between the peers. To establish this secure channel, the peers negotiate a phase 1 SA. This SA is called the IKE SA and is bidirectional because it uses the same session key for both inbound and outbound.

To authenticate each other, the peers use two methods: pre-shared key or digital signature. You can also enable an additional authentication method, XAuth, to enhance authentication.

In IKEv1, there are two possible modes in which the IKE SA negotiation can take place: main, and aggressive mode. Settings on both ends must agree; otherwise, phase 1 negotiation fails and both IPsec peers are not able to establish a secure channel.

At the end of phase 1, the negotiated IKE SA is used to negotiate the DH keys that are used in phase 2. DH uses the public key (that both ends know) plus a mathematical factor called a nonce, in order to generate a common private key. With DH, even if an attacker can listen to the messages containing the public keys, they cannot determine the secret key.

Phase 1—Network

The screenshot shows the Fortinet Phase 1 Network configuration interface. The 'Network' section is expanded, revealing various settings. A red box highlights the 'Remote gateway' and 'IP address' fields. A red arrow points from the 'Local gateway' toggle switch to a detailed view of the 'Local gateway' settings, which include 'Primary IP', 'Secondary IP', and 'Specify' options, with 'Specify' selected and the IP address set to 0.0.0.0.

Phase 1 configuration is broken down on the GUI into four sections: **Network**, **Authentication**, **Phase 1 Proposal**, and **XAUTH**. You will learn about the settings available on each section. You will learn about some of these settings in more detail on separate slides.

The section shown on this slide corresponds to the **Network** settings. The section includes the settings related to the connectivity of the IPsec tunnel:

- **IP Version:** select the IP version to use for the IPsec tunnel. Note that this defines only the IP version of the outer layer of the tunnel (after encapsulation). The packets being encapsulated (protected traffic) can be IPv4 or IPv6, and their IP version is defined in the phase 2 selectors.
- **Remote Gateway:** defines the type of the remote gateway. There are three types: **Static IP Address**, **Dialup User**, and **Dynamic DNS**. You will learn more about these types later in this lesson.
- **IP Address:** the IP address of the remote gateway. This field appears only when you select **Static IP Address** as **Remote Gateway**.
- **Interface:** refers to the interface where the IPsec tunnel terminates on the local FortiGate. Usually, this is the interface connected to the internet or the WAN. You need to make sure there is an active route to the remote gateway through this interface, otherwise the tunnel won't come up.
- **Local Gateway:** enable this setting when the interface where the tunnel terminates has multiple addresses assigned, and you want to specify which address to use for the tunnel. When you enable this setting, you see three options: **Primary IP**, **Secondary IP**, and **Specify**. Select **Specify** if you want to use an IP address different from the primary or secondary IP address.
- **Mode Config:** Enables automatic configuration through IKE. FortiGate acts as an *IKE mode config client* when you enable **Mode Config** and you set **Remote Gateway** to either **Static IP address** or **Dynamic DNS**. If you set **Remote Gateway** to **Dialup User**, FortiGate acts as an *IKE mode config server*, and more configuration options appear on the GUI. You will learn more about **Mode Config** in this lesson.

Phase 1—Network (Contd)

The screenshot shows the FortiGate GUI for configuring a network. The 'Network' section is expanded, showing various settings. A red box highlights the 'Advanced network settings' section, and a red arrow points from it to a detailed view of that section on the right.

Main Configuration Fields:

- Name: ToRemote
- Comments: Comments (0/255)
- IP version: IPv4 (selected), IPv6
- Remote gateway: Static IP address (selected), Dialup user, Dynamic DNS
- IP address: 10.200.3.1
- Interface: port1
- Local gateway: ☐
- Mode config: ☐
- NAT traversal: Enable (selected), Disable, Forced
- Keepalive frequency: 10
- Dead peer detection: ☒ On Idle, On Demand
- DPD retry count: 3
- DPD retry interval: 20 (second(s))
- Forward error correction: ☐ Egress, ☐ Ingress
- Advanced network settings: ☒ (highlighted)

Advanced network settings (Detailed View):

- Add route: ☒ Enable, ☒ Disable
- Auto discovery sender: ☒ Enable, ☒ Disable
- Auto discovery receiver: ☒ Enable, ☒ Disable
- Auto discovery forwarder: ☒ Enable, ☒ Disable
- Exchange interface IP: ☒ Enable, ☒ Disable
- Device creation: ☒ Enable, ☒ Disable
- Aggregate member: ☒ Enable, ☒ Disable

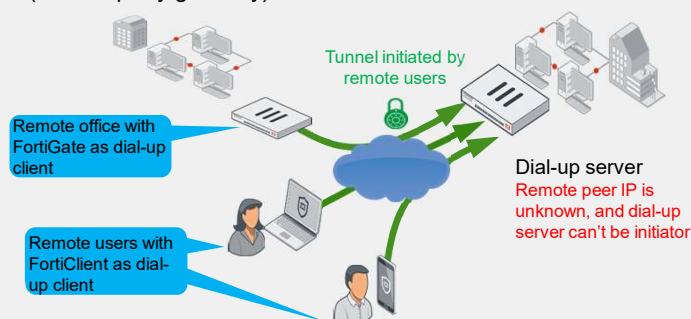
The following are the other options available on the GUI in the **Network** section:

- **NAT Traversal:** The option controls the behavior for NAT traversal. You will learn more about NAT traversal later in this lesson.
- **Keepalive Frequency:** When you enable NAT traversal, FortiGate sends keepalive probes at the configured frequency.
- **Dead Peer Detection:** Use dead peer detection (DPD) to detect dead tunnels. There are three DPD modes. **On Demand** is the default mode. You will learn more about DPD later in this lesson.
- **Forward Error Correction:** Forward error correction (FEC) is a technique that you can use to reduce the number of retransmissions in IPsec tunnels established over noisy links, at the expense of using more bandwidth. You can enable FEC on egress and ingress, and it is only supported when you disable IPsec hardware offloading. You will learn more about IPsec hardware offloading later in this lesson.
- **Advanced:**
 - **Add route:** Disable this setting if you are using a dynamic routing protocol over IPsec and do not want FortiGate to automatically add static routes.
 - **Auto discovery sender:** Enable this setting on a hub if you want the hub to facilitate ADVPN shortcut negotiation for spokes. When enabled, the hub sends a shortcut offer to the spoke to indicate that it can establish a shortcut to the remote spoke.
 - **Auto discovery receiver:** Enable this setting on a spoke if you want the spoke to negotiate an ADVPN shortcut.
 - **Exchange interface IP:** Enable this setting to allow the exchange of IPsec interface IP addresses. This allows a point-to-multipoint connection between the hub and spokes..
 - **Device creation:** Enable this setting to instruct FortiOS to create an interface for every dial-up client. To increase performance, disable this setting in dial-up servers with many dial-up clients.
 - **Aggregate member:** FortiGate allows you to aggregate multiple IPsec tunnels into a single interface. Enable this option if you want the tunnel to become an aggregate member.

Phase 1—Network—Remote Gateway

Dial-up user

- Two roles: dial-up server and client
- Dial-up server doesn't know client address
 - Dial-up client is always the initiator
- VPN peers:
 - FortiGate to FortiClient (or third-party client)
 - FortiGate to FortiGate (or third-party gateway)



You have three options when configuring the remote gateway type of your VPN: **Dialup User**, **Static IP Address**, and **Dynamic DNS**.

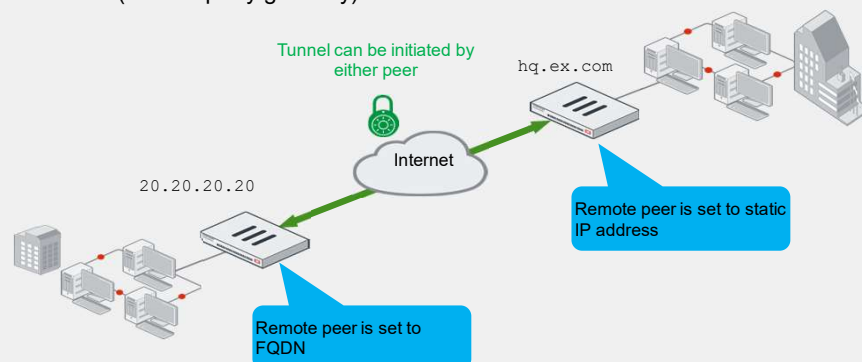
Use **Dialup User** when the remote peer IP address is unknown. The remote peer whose IP address is unknown acts as the dial-up client, and this is often the case for branch offices and mobile VPN clients that use dynamic IP addresses, and no dynamic DNS. The dial-up client must know the IP address or FQDN of the remote gateway, which acts as the dial-up server. Because the dial-up server doesn't know the remote peer address, only the dial-up client can initiate the VPN tunnel.

Usually, dial-up clients are remote and mobile employees with FortiClient on their computer or handheld devices. You can also have a FortiGate device acting as a dial-up client for a remote office. You can use one dial-up server configuration on FortiGate for multiple IPsec tunnels from many remote offices or users.

Phase 1—Network—Remote Gateway (Contd)

Static IP address/dynamic DNS

- Dynamic DNS uses FQDN
- The address of the remote peer is known
 - Local peer can be initiator or responder
- VPN peers:
 - FortiGate to FortiGate (or third-party gateway)



Use **Static IP Address** or **Dynamic DNS** when you know the remote peer address. If you select **Static IP Address**, then you must provide an IP address. If you select **Dynamic DNS**, then you must provide a fully qualified domain name (FQDN), and make sure FortiGate can resolve that FQDN. When both peers know the remote peer address, that is, the remote gateway on both peers is set to **Static IP Address** or **Dynamic DNS**, then any peer can initiate the VPN tunnel.

Note that in a dial-up setup, the dial-up client is just a VPN peer with the remote gateway set to **static IP address** or **dynamic DNS**. When setting your VPN, you can combine different types of remote gateways. For obvious reasons, a tunnel in which both peers have the remote gateway set to **Dialup user** won't work.

Phase 1—Network—IKE Mode Config

- Like DHCP, automatically configures VPN clients' virtual network settings
- By default, FortiClient VPNs use it to retrieve their VPN IP address settings from FortiGate
- You must enable **Mode Config** on both peers

IKE mode config settings are only displayed if **Remote Gateway** is set to **Dialup User**

The screenshot shows the 'Network' configuration page for IKE Mode Config. The 'Remote gateway' is set to 'Dialup user' and 'Mode config' is enabled. The 'Assign client IP' is set to 'Range' with the IP range '10.200.200.1-10.200.200.10' and subnet mask '255.255.255.255'. The DNS is set to 'Specify' with the IP '0.0.0.0'.

IKE **Mode Config** is similar to DHCP because a server assigns network settings such as IP address, netmask, and DNS servers, to clients. This assignment takes place over IKE messages.

When you enable **Mode Config** on a FortiGate device acting as dial-up server, it pushes network settings to dial-up clients. The dial-up clients are usually FortiClient peers, but they can also be FortiGate peers.

For IKE mode config to work, you must enable the feature on both peers. On FortiClient, **Mode Config** is enabled by default, but on FortiGate, you must manually enable it.

Note that the IKE **Mode Config** settings, are displayed on the GUI only when you set **Remote Gateway** to **Dialup User**. On the FortiGate device acting as dial-up client, you can select **Mode Config** on the GUI, but the additional settings are not displayed.

Phase 1—Network—NAT Traversal (NAT-T)

- ESP can't support NAT because it has no port numbers
- If **NAT Traversal** is set to **Enable**, it detects whether NAT devices exist on the path
 - If yes, both ESP and IKE use UDP port 4500
 - Recommended if the initiator or responder is behind NAT
- If **NAT Traversal** is set to **Forced**:
 - ESP and IKE always use UDP port 4500, even when there are no NAT devices on the path
- Keepalive probes are sent frequently to keep the connection across the routers active

The screenshot shows the 'Network' configuration page in FortiGate. The 'NAT traversal' section is highlighted with a red box. It includes the following settings:

- IP version:** IPv4 (selected), IPv6
- Remote gateway:** Static IP address (selected), Dialup user, Dynamic DNS
- IP address:** 10.200.3.1
- Interface:** port1
- Local gateway:** (disabled)
- Mode config:** (disabled)
- NAT traversal:** Enable (selected), Disable, Forced
- Keepalive frequency:** 10

The ESP protocol usually has problems crossing devices that are performing NAT. One of the reasons is that ESP does not use port numbers, like TCP and UDP do, to differentiate one tunnel from another.

To solve this, NAT transversal (NAT-T) was added to the IPsec specifications. When NAT-T is enabled on both ends, peers can detect any NAT device along the path. If NAT is found, then the following occurs on both peers:

- IKE negotiation switches to using UDP port 4500.
- ESP packets are encapsulated in UDP port 4500.

So, if you have two FortiGate devices that are behind, for example, an ISP modem that performs NAT, you will probably need to enable this setting.

When you set the **NAT Traversal** setting to **Forced**, UDP port 4500 is always used, even when there is no NAT device along the path.

When you enable NAT-T, the **Keepalive Frequency** option shows the interval (in seconds) at which FortiGate sends keepalive probes. You need NAT-T when there is one or more routers along the path performing NAT. The purpose of the keepalive probes is to keep the IPsec connection active across those routers along the path.

Phase 1—Network—Dead Peer Detection (DPD)

- Mechanism to detect a dead tunnel
- Useful in redundant VPNs, where multiple paths are available
- Three modes:
 - **On Demand:** DPD probes are sent when there is no inbound traffic
 - **On Idle:** DPD probes are sent when there is no traffic
 - **Disabled:** only reply to DPD probes—don't send probes

The screenshot shows the 'Network' configuration page in FortiGate. The 'Dead peer detection' section is highlighted with a red box. It includes a toggle switch for 'Dead peer detection' which is turned on, and two radio buttons for 'On Idle' and 'On Demand', with 'On Demand' being the selected mode. Other visible settings include 'IP version' set to 'IPv4', 'Remote gateway' set to 'Static IP address', 'IP address' set to '10.200.3.1', 'Interface' set to 'port1', 'Local gateway' disabled, 'Mode config' disabled, 'NAT traversal' set to 'Enable', 'Keepalive frequency' set to '10', 'DPD retry count' set to '3', and 'DPD retry interval' set to '20' seconds. There are also checkboxes for 'Forward error correction' (Egress and Ingress) which are currently unchecked.

After the peers negotiate the IPsec SAs of a tunnel and, therefore, the tunnel is considered up, the peers usually don't negotiate another IPsec SA until it expires. In most cases, the IPsec SA expires every few hours. This means that if there is a network disruption along the path of the tunnel before the IPsec SA expires, the peers will continue to send traffic through the tunnel even though the communication between the sites is disrupted.

When you enable DPD, DPD probes are sent to detect a failed (or dead) tunnel and bring it down before its IPsec SAs expire. This failure detection mechanism is very useful when you have redundant paths to the same destination, and you want to fail over to a backup connection when the primary connection fails to keep the connectivity between the sites up.

FortiGate supports three DPD modes:

- **On Demand:** FortiGate sends DPD probes if there is only outbound traffic through the tunnel, but no inbound. Because network applications are usually bidirectional, observing only traffic on the outbound direction could be an indication of a network failure.
- **On Idle:** FortiGate sends DPD probes when no traffic is observed in the tunnel. An idle tunnel does not necessarily mean the tunnel is dead. Avoid this mode if you have many tunnels, because the overhead introduced by DPD can be very resource intensive.
- **Disabled:** FortiGate replies only to DPD probes received. FortiGate never sends DPD probes to the remote peer and therefore cannot detect a dead tunnel.

The default DPD mode is **On Demand**. In terms of scalability, **On Demand** is a better option than **On Idle**.

Phase 1—Authentication

The screenshot shows the FortiGate configuration interface for Phase 1 Authentication. The main configuration area has the following settings:

- Method:** Pre-shared Key (selected), Signature
- Pre-shared key:** [Redacted]
- IKE:** Version 1, Version 2 (selected)
- Mode:** Aggressive (selected), Main (ID protection)

Two red arrows point from the highlighted settings to expanded views:

- An arrow from the **Pre-shared Key** method points to a field for the pre-shared key (masked with dots).
- An arrow from the **Version 2** IKE version points to a detailed view of the IKE configuration, which includes:
 - IKE:** Version 1, Version 2 (selected)
 - Mode:** Aggressive (selected), Main (ID protection)
 - Accepted peer ID:** Any peer ID (selected)
 - XAUTH:** [Toggle switch]
 - Accepted peer ID dropdown options:** Any peer ID, Any peer ID (highlighted in blue), Specific peer ID

Now, you will learn about the **Authentication** section in phase 1 configuration:

- **Method:** FortiGate supports two authentication methods: **Pre-shared Key** and **Signature**. When you select **Pre-shared Key**, you must configure both peers with the same pre-shared key. When you select **Signature**, phase 1 authentication is based on digital certificate signatures. Under this method, the digital signature on one peer is validated by the presence of the CA certificate installed on the other peer. That is, on the local peer, you need to install both the local peer's certificate and the CA certificate that issued the remote peer certificate.
- **Version:** allows you to select the IKE version to use. When selecting version 2, aggressive and main modes disappear because they don't apply to IKEv2.
- **Mode:** refers to the IKEv1 mode. Two options are available: **Aggressive** and **Main (ID protection)**. You will learn more about these modes in this lesson.

Phase 1—Authentication—Modes

Aggressive

- Not as secure as main mode
- Faster negotiation (three packets exchanged)
- Required when peer ID check is needed

Main

- More secure
- Slower negotiation (six packets exchanged)
- Often used when peer ID check is not needed

IKE supports two different negotiation modes: main and aggressive. Which one should you use?

To answer that question, we can analyze three categories: security, performance, and deployment.

Security wise, main mode is considered more secure because the pre-shared key hash is exchanged encrypted, whereas in aggressive mode, the hash is exchanged unencrypted. Although the attacker would still have to guess the cleartext pre-shared key for the attack to be successful, the fact that the pre-shared key hash has been encrypted in main mode reduces considerably the chances of a successful attack.

In terms of performance, aggressive mode may be a better option. This is because the negotiation is completed after only three packets are exchanged, whereas in main mode, six packets are exchanged. For this reason, you may want to use aggressive mode when a great number of tunnels terminate on the same FortiGate device, and performance is a concern.

Another use case for aggressive mode, is when there is more than one dial-up tunnel terminating on the same FortiGate IP address, and the remote peer is authenticated using a peer ID because its IP address is dynamic. Because peer ID information is sent in the first packet in an aggressive mode negotiation, then FortiGate can match the remote peer with the correct dial-up tunnel. The latter is not possible in main mode because the peer ID information is sent in the last packet, and after the tunnel has been identified.

When both peers know each other's IP address or FQDN, you may want to use main mode to take advantage of its more secure negotiation. In this case, FortiGate can identify the remote peer by its IP address and, as a result, associate it with the correct IPsec tunnel.

Phase 1—Phase 1 Proposal

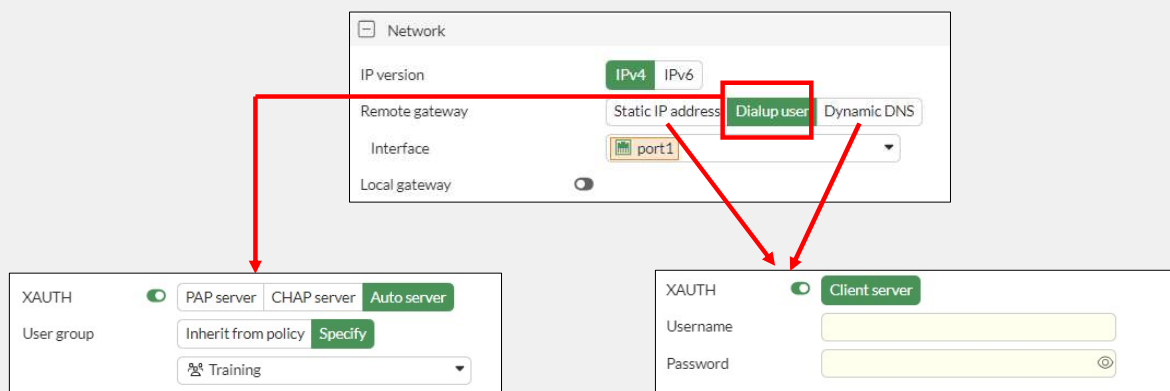
The screenshot displays the 'Phase 1 proposal' configuration window. It features a table of proposals with columns for 'Encryption - authentication', 'Diffie-Hellman groups', 'Key lifetime', and 'Local ID'. The second proposal is highlighted with a red box, indicating 'AES256' for encryption and 'SHA256' for authentication. Two red arrows point from this proposal to two smaller inset windows. The top inset shows the 'Encryption - authentication' dropdown menu with 'AES256' selected. The bottom inset shows the 'Diffie-Hellman groups' dropdown menu with 'SHA256' selected.

Now, you will learn about the **Phase 1 Proposal** section of phase 1 configuration. This section allows you to enable the different proposals that FortiGate supports when negotiating the IKE SA (or phase 1 SA). You can combine different parameters to suit your security needs. You must at least configure one combination of encryption and authentication algorithms, or several.

- **Encryption:** select the algorithm to use for encrypting and decrypting the data.
- **Authentication:** select the authentication algorithm to use for verifying the integrity and authenticity of the data.
- **Diffie-Hellman Groups:** The Diffie-Hellman (DH) algorithm is used during IKE SA negotiation. The use of DH in phase 1 is mandatory and can't be disabled. You must select at least one DH group. The higher the DH group number, the more secure the phase 1 negotiation is. However, a higher DH group number also results in a longer compute time.
- **Key Lifetime:** defines the lifetime of the IKE SA. At the end of the lifetime, a new IKE SA is negotiated.
- **Local ID:** if the peer accepts a specific peer ID, type that same peer ID in this field.

Phase 1—Extended Authentication (XAuth)

- XAuth adds stronger authentication: username + password
- You can authorize all users who belong to a specific user group or inherit it from the matching policy



Phase 1 supports two types of authentication: pre-shared keys and digital signatures. The XAuth extension, sometimes called phase 1.5, forces remote users to authenticate additionally with their credentials (username and password). So, additional authentication packets are exchanged if you enable it. What is the benefit? Stronger authentication.

When you set **Remote Gateway** to **Dialup User**, FortiGate acts as the authentication server. The **XAUTH** section shows the authentication server type options: **PAP Server**, **CHAP Server**, and **Auto Server**. In the example shown on this slide, **Auto Server** is selected, which means that FortiGate automatically detects the authentication protocol used by the client.

After you select the authentication server type, you configure how user group matching is performed. There are two options: **Inherit from policy** and **Choose**. The latter is used in the example on this slide, and allows you to select one of the user groups available on FortiGate. Note that, when you select **Choose**, you must configure a separate dial-up VPN for every group of users that require a different network access policy.

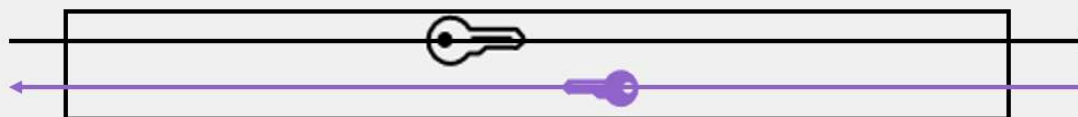
The other way to authenticate VPN users with XAuth is by selecting **Inherit from policy**. When you select this option, FortiGate authenticates users based on their matching IPsec policy and, as a result, the configuration for controlling network access is simpler. That is, you control network access by configuring multiple policies for different user groups, instead of configuring multiple tunnels for different user groups. The **Inherit from policy** option follows a similar authentication approach used for SSL VPN remote users that you learned in the SSL VPN lesson.

When **Remote Gateway** is set to **Static IP Address** or **Dynamic DNS**, FortiGate acts as the client, and the **XAUTH** section shows the **Client** option as **Type**. You can then set the credentials that FortiGate uses to authenticate against the remote peer through XAuth.

Phase 2—How it Works

- Negotiates two unidirectional IPsec SAs for ESP
 - Protected by phase 1 IKE SA

Two unidirectional SAs: one key to encrypt the outgoing traffic and another one to decrypt the incoming traffic



- When IPsec SAs are about to expire, it renegotiates
 - Optionally, if **Perfect Forward Secrecy** is enabled, FortiGate uses DH to generate new keys each time phase 2 expires
- Each phase 1 can have multiple phase 2s
 - High security subnets can have stronger ESP

After phase 1 has established a secure channel to exchange data, phase 2 begins.

Phase 2 negotiates security parameters for two IPsec SAs over the secure channel established during phase 1. ESP uses IPsec SAs to encrypt and decrypt traffic exchanged between sites, with one outbound SA and one inbound SA. Additionally, authentication header (AH) uses two integrity keys to authenticate the source of the packets and verify data integrity. AH does not provide encryption.

Phase 2 does not end when ESP begins. Phase 2 periodically renegotiates IPsec SAs to maintain security. If you enable **Perfect Forward Secrecy**, each time phase 2 expires, FortiGate uses DH to recalculate new secret keys. In this way, new keys are not derived from older keys, making it much harder for an attacker to crack the tunnel.

Each phase 1 can have multiple phase 2s. When would this happen? For example, you may want to use different encryption keys for each subnet whose traffic is crossing the tunnel. How does FortiGate select which phase 2 to use? By checking which phase 2 selector (or quick mode selector) matches the traffic.

Phase 2—Phase 2 Selectors

- Determines the encryption domain
 - You can configure multiple selectors for granular control
 - If traffic does not match a selector, it is dropped
 - In point-to-point VPNs, selectors must match
 - The source on one FortiGate is the destination setting on the other
- Select which selector to use using:
 - **Local Address** and **Remote Address**
 - **Protocol** number
 - **Local Port** and **Remote Port**

Phase 2 selectors

+ Create new Edit Delete

Search

Name	Local Address	Remote Address	Comments
ToRemote	10.0.11.0/255.255.255.0	172.20.1.0/255.255.255.0	

Local address: IP Address Subnet Address IP Range
10.0.11.0 255.255.255.0

Remote address: IP Address Subnet Address IP Range
172.20.1.0 255.255.255.0

Advanced

Local port: All Specify
Remote port: All Specify
Protocol: All Specify

In phase 2, you must define the encryption domain (or interesting traffic) of your IPsec tunnel. The encryption domain refers to the traffic that you want to protect with IPsec, and it is determined by your phase 2 selector configuration.

You can configure multiple selectors to have more granular control over traffic. When you configure a phase 2 selector, you specify the encryption domain by indicating the following network parameters:

- **Local Address** and **Remote Address**: as seen in the example shown on this slide, you can define IPv4 or IPv6 addresses using different address scopes. When selecting **Named Address** or **Named IPv6 Address**, FortiGate allows you to select an IPv4 or IPv6 firewall address object, respectively, configured in the system.
- **Protocol**: is in the **Advanced** section, and is set to **All** by default.
- **Local Port** and **Remote Port**: are also shown in the **Advanced** section, and are set to **All** by default. This applies only to port-based traffic such as TCP or UDP. You will learn more about the **Advanced** section later in this lesson.

Note that after the traffic is accepted by a firewall policy, traffic is dropped before entering the IPsec tunnel if the traffic does not match any of the phase 2 selectors configured. For this reason, usually, it's more intuitive to filter traffic with firewall policies. So, if you don't want to use phase 2 selector filtering, you can just create one phase 2 selector with both the local and remote addresses set to any subnet, like in the example shown on this slide, and then use firewall policies to control which traffic is accepted on the IPsec tunnel.

In addition, the phase 2 selector network parameters on both peers must match if the tunnel is point-to-point, that is, when the remote gateway is *not* set to dial-up user.

Phase 2—Phase 2 Proposal

- Determines the encryption algorithms
 - You can configure multiple proposals for added flexibility
 - Impacts performance and hardware offloading
- You can enable replay detection to protect against ESP replay attacks
 - Local setting

Encryption and authentication algorithms for IPsec encryption

For every phase 2 selector, you need to configure one or more phase 2 proposals. A phase 2 proposal defines the algorithms supported by the peer for encrypting and decrypting the data over the tunnel. You can configure multiple proposals to offer more options to the remote peer when negotiating the IPsec SAs.

Like in phase 1, you need to select a combination of encryption and authentication algorithms. Some algorithms are considered more secure than others, so make sure to select the algorithms that conform with your security policy. However, note that the selection of the algorithms has a direct impact on FortiGate IPsec performance. For example, **3DES** is known to be a much more resource-intensive encryption algorithm than **DES** and **AES**, which means that your IPsec throughput could be negatively impacted if you select **3DES** as the encryption algorithm. Also, note that if you select **NULL** as the encryption algorithm, traffic is not encrypted.

In addition, some encryption algorithms, such as **CHACHA20POLY1305**, are not supported for hardware offload. That is, if you have a FortiGate device that contains network processor (NP) units, you can achieve higher IPsec performance if you select an algorithm that is supported for IPsec offload by your NP unit model, such as AES or DES. For a list of supported encryption algorithms for IPsec hardware offloading, refer to <https://docs.fortinet.com>.

When configuring the phase 2 proposal, you can select **Enable Replay Detection** to detect antireplay attacks on ESP packets. Note that this is a local setting and, therefore, it is not included as part of the proposals presented by the peer during phase 2 negotiation.

Also, if you enable **Perfect Forward Secrecy**, FortiGate uses DH to enhance security during the negotiation of IPsec SAs.

Phase 2—Phase 2 Proposal (Contd)

- IPsec SA expires based on the number of:
 - Seconds** (time-based)
 - Kilobytes** (volume-based)
 - Both** (whichever expires first)
- Key lifetime thresholds do not have to match for tunnel to come up
- Auto-negotiate** prevents disruption caused by SA renegotiation
- Autokey Keep Alive** keeps the tunnel up

Auto-negotiate ☒ Enable ☐ Disable

Autokey keep alive ☒ Enable ☐ Disable

These settings control when SA renegotiation occurs

Advanced

Encryption - authentication

AES128	-	SHA1	X
AES256	-	SHA1	X
AES128	-	SHA256	X
AES256	-	SHA256	X
AES128GCM			X
AES256GCM			X
CHACHA20POLY1305			X

Replay detection ☒ Enable ☐ Disable

Perfect forward secrecy (PFS) ☒ Enable ☐ Disable

Diffie-Hellman groups

☐ 1	☐ 15	☒ 5
☒ 14	☐ 16	☐ 19
☐ 17	☐ 18	☐ 27
☐ 20	☐ 21	☐ 30
☐ 28	☐ 29	
☐ 31	☐ 32	

Local port ☒ All ☐ Specify

Remote port ☒ All ☐ Specify

Protocol ☒ All ☐ Specify

Auto-negotiate ☒ Enable ☐ Disable

Autokey keep alive ☒ Enable ☐ Disable

Key lifetime

☒ Seconds	☐ Kilobytes	☐ Both
---	------------------------------------	-------------------------------

43200 second(s)

IPsec SAs are periodically renegotiated to improve security, but when does that happen? It depends on the key lifetime settings configured on the phase 2 proposal.

The expiration of an IPsec SA is determined by the lifetime type and threshold configured. By default, **Key Lifetime** is set to **Seconds** (time-based). This means that when the SA duration reaches the number of seconds set as **Seconds**, the SA is considered expired. You can also set the key lifetime to **Kilobytes** (volume-based), upon which the SA expires after the amount of traffic encrypted and decrypted using that SA reaches the threshold set. Alternatively, you can select **Both** as the key lifetime type, upon which FortiGate tracks both the duration of the SA and the amount of traffic. Then, when any of the two thresholds is reached, the SA is considered expired. Note that the key lifetime thresholds do not have to match for the tunnel to come up. When thresholds are different, the peers agree on using the lowest threshold value offered between the two.

When IPsec SAs expire, FortiGate needs to negotiate new SAs to continue sending and receiving traffic over the IPsec tunnel. Technically, FortiGate deletes the expired SAs from the respective phase 2 selectors, and installs new ones. If IPsec SA renegotiation takes too much time, then FortiGate might drop interesting traffic because of the absence of active SAs. To prevent this, you can enable **Auto-negotiate**. When you do this, FortiGate not only negotiates new SAs before the current SAs expire, but it also starts using the new SAs right away. The latter prevents traffic disruption by IPsec SA renegotiation.

Another benefit of enabling **Auto-negotiate** is that the tunnel comes up and stays up automatically, even when there is no interesting traffic. When you enable **Autokey Keep Alive** and keep **Auto-negotiate** disabled, the tunnel does not come up automatically unless there is interesting traffic. However, after the tunnel is up, it stays that way because FortiGate periodically sends keep alive packets over the tunnel. Note that when you enable **Auto-negotiate**, **Autokey Keep Alive** is implicitly enabled.

IPsec Hardware Offloading

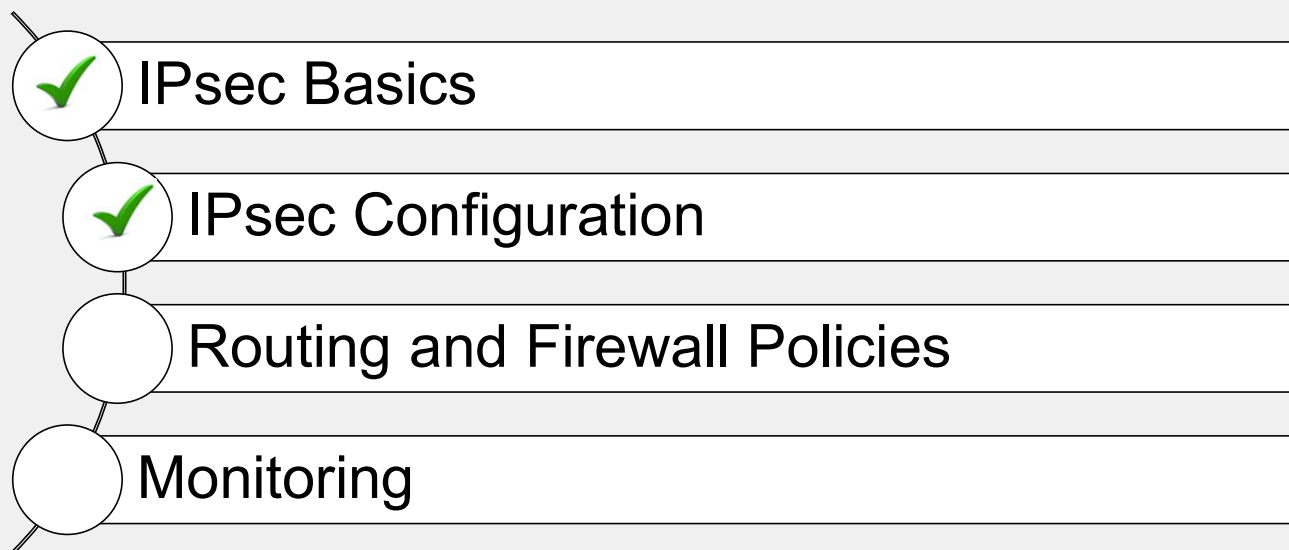
- On some FortiGate models, you can offload IPsec encryption and decryption to hardware
- Hardware offloading capabilities and supported algorithms vary by processor type and model
- By default, offloading is enabled for supported algorithms
 - You can manually disable offloading:

```
config vpn ipsec phase1-interface
    edit ToRemote
        set npu-offload disable
    next
end
```

On some FortiGate models, you can offload the encryption and decryption of IPsec traffic to hardware. The algorithms that are supported depend on the NP unit model present on FortiGate. For a list of supported encryption algorithms for IPsec hardware offloading, refer to <https://docs.fortinet.com>.

By default, hardware offloading is enabled for the supported algorithms. This slide shows the commands you can use to disable hardware offloading per tunnel, if necessary.

Lesson Progress



Good job! You now understand IPsec configuration.

Now, you will learn about routing and firewall policies for IPsec traffic.

Routing and Firewall Policies

Objectives

- Understand route-based IPsec VPNs
- Configure firewall policies for IPsec traffic
- Configure a redundant VPN between two FortiGate devices

After completing this lesson, you should be able to achieve the objectives shown on this slide.

By demonstrating competence in routing and firewall policies for IPsec VPNs, you will be able to set up appropriate routing and firewall policies on FortiGate, and add redundancy to your IPsec VPN deployment.

Route-Based IPsec VPNs

- Types of IPsec VPNs:
 - Route-based
 - Virtual interface for each VPN: VPN matching based on routing
 - Policy-based
 - Legacy: VPN matching based on policy. Not recommended.
- Route-based VPNs benefits:
 - Simpler operation and configuration
 - Redundancy
 - Support for:
 - L2TP-over-IPsec
 - GRE-over-IPsec
 - Dynamic routing protocols

FortiGate supports two types of IPsec VPNs: route-based and policy-based. Policy-based is a legacy IPsec VPN that is supported only for backward compatibility reasons, and its use *is not recommended* for new deployments. Unless otherwise stated, all IPsec VPN references in this lesson are for route-based IPsec VPNs.

In a route-based IPsec VPN, FortiGate automatically adds a virtual interface with the VPN name. This means that not only can you configure routing and firewall policies for IPsec traffic in the same way you do for non-IPsec traffic, but you also can leverage the presence of multiple connections to the same destination to achieve redundancy.

Another benefit of route-based IPsec VPNs is that you can deploy variations of IPsec VPNs such as L2TP-over-IPsec and GRE-over-IPsec. In addition, you can also enable dynamic routing protocols for scalability purposes and best path selection.

Routes for IPsec VPNs

Dial-up user

```
config vpn ipsec phase1-interface
edit "Dialup"
    set add-route enable | disable
next
end
```

- **add-route is enabled (default)**
 - No need to configure static routes
 - Static routes are added after phase 2 is up
 - The destination is the local network presented by the dial-up client during phase 2 negotiation
 - The default route distance is 15
 - Static routes are deleted after phase 2 is down
- **add-route is disabled**
 - Useful when dynamic routing protocol is used
 - Dynamic routing protocol takes care of routing updates

Static IP address / dynamic DNS

- Static routes are needed

Network > Static Routes

Edit Static Route

Destination ⓘ Subnet Internet Service
 10.0.2.0/255.255.255.0

Interface ToRemote + ✕

Administrative Distance ⓘ 10

Comments Write a comment... 0/255

Status Enabled Disabled

+ Advanced Options

Select virtual interface of IPsec VPN

Although you can use dynamic routing protocols for IPsec VPNs, this lesson covers only the use of static routes.

The routing configuration needed for your IPsec VPN depends on the type of remote gateway configured. When you set the remote gateway to **Dialup User** and enable `add-route`, FortiGate automatically adds a static route for the local network presented by the remote peer during phase 2 negotiation. In addition, the route is added to the routing table only after phase 2 is up. If phase 2 goes down, the static route is removed from the routing table.

When you set the remote gateway to **Dialup User** and disable `add-route`, FortiGate does not add static routes automatically. In this case, a dynamic routing protocol is used between the remote peers to exchange routing information.

When the remote gateway is set to **Static IP Address** or **Dynamic DNS**, you must configure static routes. When you configure a static route, you select the virtual interface of the IPsec tunnel as the outgoing interface.

Firewall Policies for IPsec VPNs

- At least one firewall policy is needed for a tunnel to come up
- Usually two firewall policies are configured for every tunnel

Policy & Objects > Firewall Policy

Name: Remote_out

Schedule: always

Action: ☒ ACCEPT ☐ DENY

Incoming interface: port3

Outgoing interface: ToRemote

Source & Destination: Show logic

Source: LOCAL_SUBNET

User/group: +

Destination: REMOTE_SUBNET

Service: ALL

Firewall/Network Options

NAT: ☒

Virtual interface matches phase 1 name

Allow and inspect the traffic coming from/going to the IPsec virtual interface

Policy & Objects > Firewall Policy

Name: Remote_In

Schedule: always

Action: ☒ ACCEPT ☐ DENY

Incoming interface: ToRemote

Outgoing interface: port3

Source & Destination: Show logic

Source: REMOTE_SUBNET

User/group: +

Destination: LOCAL_SUBNET

Service: ALL

Firewall/Network Options

NAT: ☒

You must configure at least one firewall policy that accepts traffic on your IPsec tunnel. Otherwise, the tunnel will not come up.

When you configure firewall policies for non-IPsec traffic, the policy determines the direction of the traffic that initiates sessions. The same applies to IPsec traffic. For this reason, you usually want to configure at least two firewall policies for your IPsec VPN: one incoming policy and one outgoing policy. The incoming policy allows traffic initiated from the remote site, while the outgoing policy allows traffic to be initiated from the local network.

Note that the policies are configured with the virtual tunnel interface (or phase 1 name) as the incoming or outgoing interface.

Redundant VPNs

- If the primary VPN tunnel fails, FortiGate then routes traffic through the backup VPN
- *Partially* redundant: *one* peer has two connections



- *Fully* redundant: *both* peers have two connections



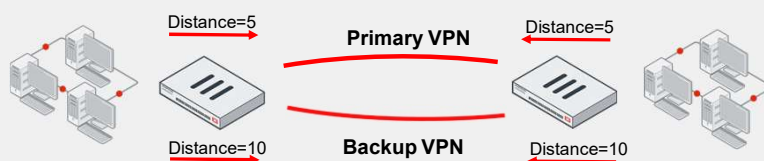
How can you make your IPsec VPN deployment more resilient? Provide a second ISP connection to your site and configure two IPsec VPNs. If the primary IPsec VPN fails, another tunnel can be used instead.

There are two types of redundant VPNs:

- Partially redundant: on one peer (usually the hub, where a backup ISP is available if the main ISP is down), each VPN terminates on *different* physical ports. That way, FortiGate can use an alternative VPN. On the other peer, each VPN terminates on the *same* physical port—so the spoke is not fault tolerant.
- Fully-redundant: both peers terminate their VPNs on different physical ports, so they are both fault tolerant.

Redundant VPN Configuration

- Add one phase 1 configuration for each tunnel. You should enable DPD on both ends.
- Add at least one phase 2 definition for each phase 1
- Add one static route for each path
 - Use distance or priority to select primary routes over backup routes
 - Alternatively, use dynamic routing
- Configure firewall policies for each IPsec interface



So, how do you configure a partially or fully redundant VPN?

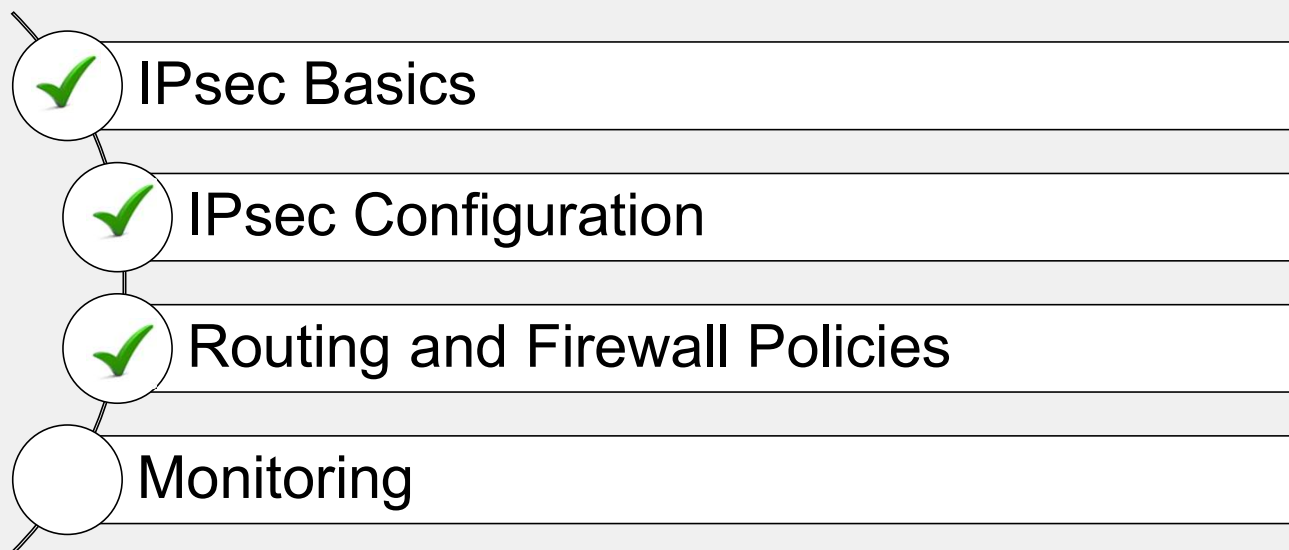
First, create one phase 1 for each path—one phase 1 for the primary VPN and one for the backup VPN. You should also enable DPD on both ends.

Second, create at least one phase 2 definition for each phase 1.

Third, you must add at least one static route for each VPN. Routes for the primary VPN must have a lower distance (or lower priority) than the backup. This causes FortiGate to use the primary VPN while it's available. If the primary VPN fails, then FortiGate automatically uses the backup route. Alternatively, you could use a dynamic routing protocol, such as OSPF or BGP.

Finally, configure firewall policies to allow traffic through both the primary and backup VPNs.

Lesson Progress



Good job! You now understand routing and firewall policies for IPsec traffic.

Now, you will learn about monitoring IPsec VPNs and reviewing their logs.

Monitoring

Objectives

- Monitor IPsec VPNs and review logs
- Troubleshoot IPsec VPN issues

After completing this section, you should be able to achieve the objectives shown on this slide.

By demonstrating competence in monitoring and logs, you will be able to monitor IPsec VPN and investigate common issues.

IPsec VPN Status—IPsec Monitor Widget

- Monitor IPsec VPN tunnels
 - Display status and statistics
 - Bring up or bring down VPNs

Dashboard > Network > IPsec

VPN status

Bring down the entire tunnel or the phase 2 only

Data received

Data sent

Phase 1 name and status

Phase 2 name and status

More columns available

On the GUI dashboard, you can use the IPsec widget to monitor the status of your IPsec VPNs. The widget shows the phase 1 and phase 2 status of an IPsec VPN.

You can also bring up or bring down individual VPNs, and get additional details. When you bring up an IPsec VPN using the IPsec widget, you can choose between bringing up a particular phase 2 selector or all phase 2 selectors in that VPN. Because bringing up a phase 2 selector requires bringing up its phase 1 first, then bringing up a phase 2 selector results in its phase 1 also coming up.

To bring down the VPN, you can choose between bringing down a particular phase 2 selector, all selectors, or the entire tunnel. When you bring down the entire tunnel, you bring down all phase 2 selectors as well as the phase 1.

The **Name** column indicates the VPN status. The VPN is up when at least one of its phase 2 selectors is up. If all phase 2 selectors are down, the VPN status is also down. The **Phase 1** and **Phase 2 Selectors** columns indicate the status of phase 1 and phase 2 selectors, respectively.

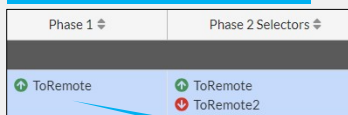
The IPsec widget also displays the amount of data sent and received through the tunnel. When you right-click any of the columns, a menu opens with a list of all the columns available. You can enable additional columns to get further details about the IPsec tunnels.

In the example shown on this slide, the **ToRemote** VPN is up because at least one of its phase 2 selectors (**ToRemote**) is up.

Monitor IPsec Routes

- IPsec routes appear in the routing table after:
 - Phase 1 comes up, if the remote gateway is set to static IP address or dynamic DNS

Dashboard > Network > IPsec



Phase 1 is up

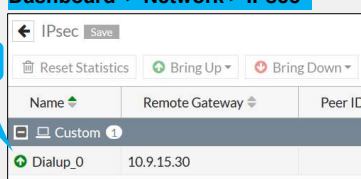
Dashboard > Network > Static & Dynamic Routing

Network	Gateway IP	Interfaces	Distance
0.0.0.0/0	10.200.1.254	port1	10
10.0.1.0/24	0.0.0.0	port3	0
10.0.2.0/24		ToRemote	10

- Phase 2 comes up, if the remote gateway is set to dial-up user

Dashboard > Network > IPsec

Phase 2 is up



Dashboard > Network > Static & Dynamic Routing

Network	Gateway IP	Interfaces	Distance
0.0.0.0/0	10.9.15.254	port1	10
10.0.2.0/24	10.9.15.30	Dialup	15

If you set the remote gateway to **Static IP Address** or **Dynamic DNS**, the static routes for these tunnels become active in the routing table after phase 1 comes up. Phase 1 negotiation is started automatically because automatic negotiation is enabled on phase 1 by default. This behavior allows FortiGate to match interesting traffic to the right tunnel. Moreover, if phase 2 is not up, traffic matching the static route triggers a phase 2 negotiation, which eventually results in the tunnel (or phase 2) to come up.

When you set the remote gateway to **Dialup User**, by default, a static route for the destination network is added after phase 2 comes up. The distance set for the static route is 15. If phase 2 goes down, the route is removed from the routing table.

IPsec Logs

Log & Report > System Events > VPN Events

Phase 2 is up (tunnel is up)

Double-click any log to get more details

Date/Time	Level	Action	Status	Message	VPN Tunnel
2024/09/13 06:24:16	Notice	negotiate	success	progress IPsec phase 2	ToRemote
2024/09/13 06:24:16	Notice	negotiate	success	negotiate IPsec phase 2	ToRemote
2024/09/13 06:24:16	Notice	negotiate	success	progress IPsec phase 2	ToRemote
2023/09/13 06:24:16	Notice	tunnel-up		IPsec connection status change	ToRemote
2024/09/13 06:24:16	Notice	phase2-up		IPsec phase 2 status change	ToRemote
2024/09/13 06:24:16	Notice	install_sa		install IPsec SA	ToRemote
2024/09/13 06:24:16	Notice	negotiate	success	progress IPsec phase 2	ToRemote
2024/09/13 06:24:08	Notice	negotiate	success	progress IPsec phase 1	ToRemote
2024/09/13 06:24:08	Notice	negotiate	success	progress IPsec phase 1	ToRemote
2024/09/13 06:24:07	Notice	delete_phase1_sa		delete IPsec phase 1 SA	ToRemote
2024/09/13 06:24:07	Notice	phase2-down		IPsec phase 2 status change	ToRemote
2024/09/13 06:24:07	Notice	tunnel-down		IPsec connection status change	ToRemote

Log Details

General

Absolute Date/Time: 2024-09-13
 Last Access Time: 06:24:08
 VDOM: root
 Log Description: Progress IPsec phase 1

Source

Local IP: 10.200.1.1
 Source Country/Region: Reserved
 FortiClient ID: N/A
 User: Remote-FortiGate
 Group: N/A
 XAUTH User: N/A
 XAUTH Group: N/A

Action

Action: negotiate
 Status: success
 Result: DONE

Phase 1 is DONE (up)

FortiGate logs IPsec VPN events by default. To view IPsec VPN event logs, click **Log & Report > System Events > VPN Events**.

The logs track the progress of phase 1 and phase 2 negotiations, and report on tunnel up and down events and DPD failures, among other events. For more information about IPsec logs, visit <https://docs.fortinet.com>.

IPsec SA Management

```
# diagnose vpn tunnel ?
```

```
down          Shut down tunnel
up            Activate tunnel
list          list all tunnel
flush         Flush tunnel SAs.
...
```

The same command `diagnose vpn tunnel` offers options for listing, shutting down, activating, or flushing a VPN tunnel.

IPsec SA

```
# diagnose vpn tunnel list name Hub2Spoke1
list IPsec tunnel by names in vd 0
-----
name=Hub2Spoke1 ver=1 serial=2 10.10.1.1:0->10.10.2.2:0
bound_if=6 lgwy=static/1 tun=intf/0 mode=auto/1 encap=none/8 options[0008]=npu
proxyid_num=1 child_num=0 refcnt=8 ilast=11 olast=3 auto-discovery=0
stat: rxp=513 txp=129 rxb=459050 txb=93
dpd: mode=on-demand on=1 idle=20000ms retry=3 count=0 seqno=36
natt: mode=none draft=0 interval=0 remote_port=0
proxyid=Hub2Spoke1 proto=0 sa=1 ref=2 serial=1
src: 0:192.168.1.0/255.255.255.0:0
dst: 0:10.10.20.0/255.255.255.0:0
SA: ref=7 options=2e type=00 soft=0 mtu=1438 expire=41195/0B replaywin=1024 seqno=9d esn=0
replaywin lastseq=00000200
life: type=01 bytes=0/0 timeout=43150/43200
dec: spi=01e54b14 esp=aes key=16 914dc5d092667ed436ea7f6efb867976
ah=sha1 key=20 a81b019d4cdfda32ce51e6b01d0b1ea42a74adce
enc: spi=3dd3545f esp=aes key=16 017b8ff6c4ba21eac99b22380b7de74d
ah=sha1 key=20 edd8141f4956140eef703d9042621d3dbf5cd961
dec:pkts/bytes=513/458986, enc:pkts/bytes=250/26848
npu_flag=03 npu_rgwy=10.10.2.2 npu_lgwy=10.10.1.1 npu_selid=1
```

Lists specified tunnel
information only

DPD information

Anti-replay is enabled

SA information

Hardware offload
information

The command `diagnose vpn tunnel list` displays the current IPsec SA information for all active tunnels.

The command `diagnose vpn tunnel list name <tunnel name>` provides SA information about a specific tunnel.

IPsec Tunnel Details

```
Hub # get vpn ipsec tunnel details
```

```
gateway
```

```
name: 'Hub2Spoke1'
```

```
type: route-based
```

```
local-gateway: 10.10.1.1:0 (static)
```

```
remote-gateway: 10.10.2.2:0 (static)
```

```
mode: ike-v1
```

```
interface: 'wan2' (6)
```

```
rx packets: 1025 bytes: 524402 errors: 0
```

```
tx packets: 641 bytes: 93 errors: 0
```

```
dpd: on-demand/negotiated idle: 20000ms retry: 3 count: 0
```

```
selectors
```

```
name: 'Hub2Spoke1'
```

```
auto-negotiate: disable
```

```
mode: tunnel
```

```
src: 0:192.168.1.0/0.0.0.0:0
```

```
dst: 0:10.10.20.0/0.0.0.0:0
```

```
SA
```

```
lifetime/rekey: 43200/32137
```

```
mtu: 1438
```

```
tx-esp-seq: 2ce
```

```
replay: enabled
```

```
inbound
```

```
spi: 01e54b14
```

```
enc: aes-cb 914dc5d092667ed436ea7f6efb867976
```

```
auth: sha1 a81b019d4cdfda32ce51e6b01d0b1ea42a74adce
```

```
outbound
```

```
spi: 3dd3545f
```

```
enc: aes-cb 017b8ff6c4ba21eac99b22380b7de74d
```

```
auth: sha1 edd8141f4956140eef703d9042621d3dbf5cd961
```

```
NPU acceleration: encryption(outbound) decryption(inbound)
```

Phase 1 details

Quick mode selectors

Tunnel MTU

Phase 2 SAs for each direction

Hardware acceleration

The command `get vpn ipsec tunnel details` provides information for the active IPsec tunnels.

The output shows traffic counters, negotiated quick mode selectors, and negotiated encryption, authentication, and keys.

IKE Gateway List

```
Hub # diagnose vpn ike gateway list name Hub2Spoke1
vd: root/0
name: Hub2Spoke1
version: 1
interface: wan2 6
addr: 10.10.1.1:500 -> 10.10.2.2:500
created: 3196s ago
auto-discovery: 0
IKE SA: created 1/1 established 1/1 time 6020/6020/6020 ms
IPsec SA: created 1/1 established 1/1 time 40/40/40 ms
```

When phase 1 was created

```
id/spi: 87 16b474c1ae9de3ca/67e428c8c7118617
```

```
direction: initiator
```

Is this gateway an initiator or responder?

```
status: established 3196-3190s ago = 6020ms
```

```
proposal: aes128-sha256
```

```
key: 34641b135ceeb2cd-c44a41d15dec439c
```

```
lifetime/rekey: 86400/82909
```

```
DPD sent/rcv: 00000040/0000002e
```

Clear phase 1

```
Hub # diagnose vpn ike gateway clear <name>
```

The command `diagnose vpn ike gateway list` also provides some details about a tunnel.

The command `diagnose vpn ike gateway clear` closes a phase 1. Be careful when using this command because it has a global effect. This means that running it without specifying the phase 1 name results in all phase 1s of all VDOMs being cleared.

Common IPsec Problems

Problem	Output of IKE debug	Common causes	Common solutions
Tunnel is not coming up	Error: negotiation failure	IPsec configuration mismatch	Verify phase 1 and phase 2 configurations between both peers
	Error: no SA proposal chosen	IPsec configuration mismatch	Verify phase 1 and phase 2 configurations between both peers Enable NAT-Traversal
Tunnel is unstable	DPD packet lost	ISP issue	Check internet connection
			Enable NAT-Traversal
Tunnel is up but traffic doesn't pass through it	Error in debug flow: no matching IPsec selector, drop	Traffic not matching quick mode selector	Verify quick mode selectors are correct
		NAT is enabled	Disable NAT on the VPN firewall policy
	Routing issue	Route missing or pointing to wrong device	Verify route is correctly defined Enable NAT-Traversal

This slide shows a summary of the most common IPsec problems and solutions.

If the tunnel doesn't come up, use the IKE real-time debug. In such cases, an error message usually appears.

When the tunnel is unstable, you usually see that DPD packets are being lost, which indicates that the problem might be on the ISP side.

If the tunnel is up but traffic isn't passing through it, use the debug flow. One of the peers might be dropping packets or routing traffic incorrectly. Another possibility is that the packets don't match the quick mode selectors, so FortiGate drops the packets.

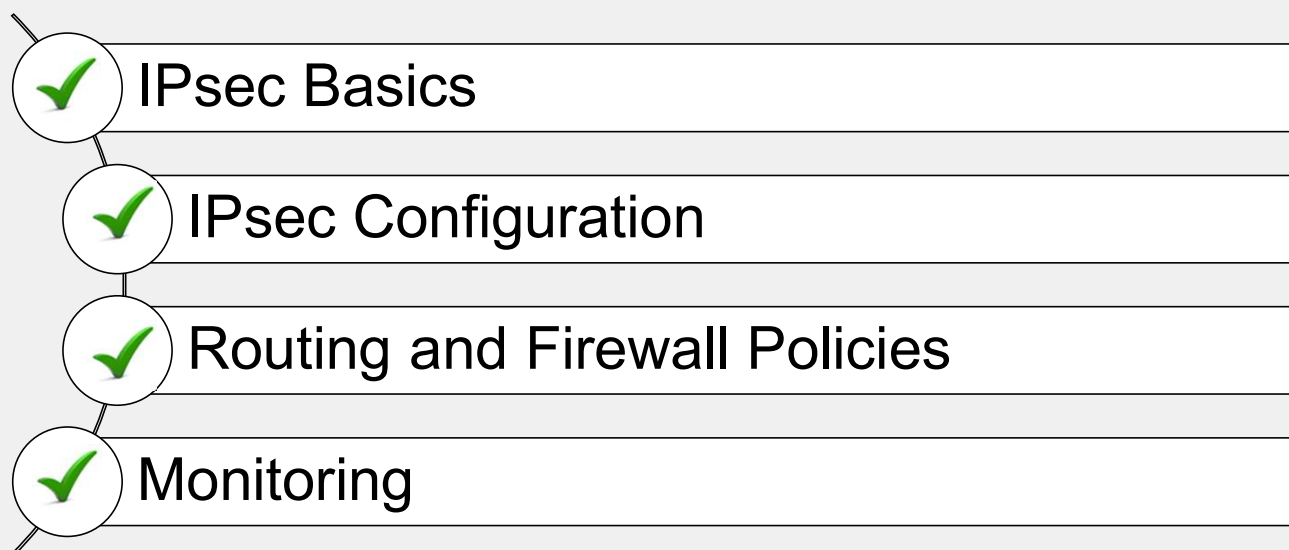
Knowledge Check

1. What is a configuration requirement for an IPsec tunnel to come up?
 - ✓ A. A firewall policy accepting traffic on the IPsec tunnel
 - B. A route for IPsec traffic

2. Which setting determines whether a tunnel is used as primary or backup without ECMP?
 - ✓ A. Routing
 - B. Firewall policies

3. When the remote gateway is set to dial-up user, a static route to the remote network is added to the routing table after _____.
 - A. Phase 1 comes up
 - ✓ B. Phase 2 comes up

Lesson Progress



Congratulations! You have completed this lesson.

Now, you will review the objectives covered in this lesson.

Review

- ✓ Configure IPsec VPN manually
- ✓ Configure IPsec VPN using the IPsec wizard
- ✓ Configure a redundant VPN between two FortiGate devices
- ✓ Monitor IPsec VPNs and review logs
- ✓ Troubleshoot IPsec VPN issues

This slide shows the objectives that you covered in this lesson.

By mastering the objectives covered in this lesson, you learned how the IPsec protocol works, and how to configure and monitor IPsec VPNs on FortiGate.